

Yufeng (Alex) Xia

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PROFILE

- MSc by Research student at the Institute for Computing Systems Architecture (ICSA), University of Edinburgh, supervised by Prof. Mahesh K. Marina. My interest lies at the intersection of **distributed systems, machine learning systems, and performance engineering**. My work spans **accelerator sharing and scheduling**, where stateful ML workloads run on contended, reclaimable GPU capacity alongside a latency-critical tenant, as well as **large-scale emulation and profiling**, the infrastructure used to measure how such systems behave at cluster scale.

I work on the layer that decides which workload gets which accelerator, and on what happens when that decision changes: preemption tolerance, state preservation, and graded degradation under resource pressure. These systems have led to publications at top-tier venues including ACM MobiCom, and what interests me most now is applying the same scheduling and state-management questions to large-scale AI inference infrastructure. **Available full time from September 2026.**

EDUCATION

- **University of Edinburgh — MSc by Research, Computer Science** Edinburgh, UK
Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN Sep 2025 – Sep 2026 (expected)
- **University of Edinburgh — BSc Computer Science** Edinburgh, UK
*Grade: **First Class Honours*** Sep 2021 – May 2025
 - **Relevant courses:** Machine Learning Systems, Machine Learning Practical, Computer Communications and Networks.

TECHNICAL SKILLS

- **Languages** C, C++, Python, Java, Bash.
- **Distributed & Systems** GPU cluster scheduling and resource allocation, preemption and state preservation, multi-node deployment on public cloud (Microsoft Azure, CloudLab), Linux, Docker, Kubernetes, Git.
- **ML Systems** PyTorch, distributed foundation-model training, edge and heterogeneous-device training systems, workload scheduling under contention.
- **Performance & Profiling** CUDA GPU programming and profiling, accelerator interference characterisation, LLVM compiler optimisation and quantisation, large-scale benchmarking and evaluation design.

RESEARCH AND PROJECT EXPERIENCE

- **Weaver — Foundation Model Training over Shared AI Compute Infrastructure** Under Submission
Research Assistant; spare-compute controller evaluation and large-scale experiments 2024 – 2026
 - **Accelerator harvesting under a latency-critical co-tenant:** Evaluated the controller that decides how much accelerator capacity an AI workload may harvest from infrastructure shared with a latency-critical tenant, a near-real-time resource-allocation problem on GPU-accelerated hardware. On a multi-site testbed it harvests up to **83%** of the available spare compute without degrading the co-tenant's performance.
 - **Profiling and large-scale training runs:** Ran GPU profiling and low-level experiments to characterise interference between training workloads and the co-located latency-critical workload, and carried out the large-scale distributed foundation-model training runs used in the evaluation.
- **Flux-DT — Elastic, Preemption-Tolerant Runtime for Stateful Workloads on Spare Compute** MSc by Research Thesis
University of Edinburgh — sole author Sep 2025 – Present
 - **State that must survive preemption:** Existing compute-sharing designs assume the harvesting workload is stateless and disposable. Built a runtime for a workload that is neither: it holds scene state that must survive interruption, so reclaiming its compute cannot simply mean killing it. The runtime lets such a workload run on bursty, reclaimable capacity without violating the priority tenant's guarantees or corrupting its own state.

- **Near-real-time control policy:** Designed and implemented a control loop that answers resource pressure with a graded set of actions — relocation, fidelity reduction, suspend and resume — rather than a single eviction decision, on the timescale the priority workload imposes.
- **Scheduling heuristic and shedding order:** Formulated work placement as a joint objective across interfering units and **designed a greedy heuristic** to decide which work is worth doing; the same ranking determines the shedding order when compute is reclaimed mid-solve.
- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
*Research Assistant; **Best Paper Award*** *Oct 2025 – Mar 2026*
 - **Measurement-driven emulation of distributed training:** Contributed to a measurement-driven emulator for evaluating distributed ML training across heterogeneous devices, supporting up to 1,800 emulated devices on a single GPU server with low latency and thermal modelling error.
- **EmuRAN — Scalable, Cost-Effective Cloud-based Emulation System** MobiCom 2026
*Research Assistant; **owned the end-to-end evaluation*** *Conditionally Accepted*
 - **Cloud deployment at scale:** Deployed and operated the emulator on public cloud infrastructure across 130 instances and 6,400 CPU cores, emulating over 10,000 devices, two orders of magnitude larger in scale than existing software solutions.
 - **Fidelity and scalability evaluation:** Designed and ran the evaluation campaign: fidelity against real hardware and scalability under node count and constrained compute, establishing that the system behaves correctly rather than merely running at scale.
- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
Research Assistant, ICSA; first author *Apr 2024 – May 2025*
 - **Compiler-level optimisation:** Built an LLVM auto-quantization framework driven by information a compiler cannot recover from source alone, reducing code size and execution time on embedded targets; low-level C/C++ and cross-compilation toolchain work throughout.
 - **Measurement on real hardware:** Evaluated the framework on embedded ARM devices, measuring power, performance and area improvements.

INDUSTRY EXPERIENCE

- **Senior Tech — Back-end Development Intern** Suzhou, China
Production backend APIs in a commercial engineering team; live platform migration to Spring Boot, code review and release testing. Java, MySQL, Redis, Kafka. *Jun 2023 – Sep 2023*

SELECTED PUBLICATIONS

- EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** *MobiCom'26 (cond. accepted)*
*Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic*
- Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** *under submission*
*Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina*
- Morphling: Emulator for Distributed Machine Learning at the Edge** *EdgeSys'26 (Best Paper)*
*Leyang Xue, **Yufeng Xia**, Ismaeel Bashir, Eren Mendi, Jiaxun Yang, Myungjin Lee, Mahesh Marina*
- Wasp: Foundation Model Training System for the Edge** *SEC'26*
*Leyang Xue, **Yufeng Xia**, Myungjin Lee, Mahesh Marina*
- CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization** *under submission*
***Yufeng Xia**, Pei Mu, Antonio Barbalace*

AWARDS, LANGUAGES AND SERVICE

- **Award Best Paper**, EdgeSys 2026 (co-located with ACM MobiSys 2026), for *Morphling*.
- **Languages** Mandarin (native), English (fluent).
- **Service** Student volunteer, EuroSys 2026 (on-site conference operations).